

ORIGINAL ARTICLE

Choroidal melanoma patient views on the importance of treatment characteristics and outcomes, and physicians' information provision in the Netherlands

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Abstract

Purpose: To identify the treatment characteristics and outcomes that Dutch patients with large or juxtapapillary choroidal melanoma consider important when choosing between enucleation and proton beam therapy (PBT). We further determined to what extent these treatment characteristics and outcomes are discussed with newly diagnosed choroidal melanoma patients.

Methods: Through a literature search, stakeholder interviews and feedback from patients, we identified 15 treatment characteristics and outcomes that patients might consider important for deciding between enucleation and PBT. Using a ranking survey, we selected the seven most important characteristics. Next, we conducted a Swing Weighting (SW) survey among treated and newly diagnosed large or juxtapapillary choroidal melanoma patients to determine the relative weight patients assign to these characteristics for treatment decision-making. Furthermore, we assessed which characteristics and outcomes are discussed in current practice with newly diagnosed choroidal melanoma patients eligible for both enucleation and PBT using audio recordings of consultations.

Results: One hundred sixty-five participants completed the SW survey. The patients found the risk of tumour recurrence and eye loss most important. These were mentioned in 47% and 42% of the 110 audio-recorded consultations, respectively. The two most-often mentioned characteristics were treatment course (96%) and decrease in visual acuity (85%).

Conclusion: Tumour local control and eye preservation matter the most to newly diagnosed choroidal melanoma patients when facing a treatment decision. However, ocular oncologists do not routinely provide this information to patients. The results can be used to align counselling of newly diagnosed patients with choroidal melanoma with what matters most to patients.

KEYWORDS

choroidal melanoma, enucleation, information provision, proton beam therapy, treatment decision-making

1 | INTRODUCTION

Choroidal melanoma is a rare malignancy in adults; yet it is the most frequently occurring ocular malignancy with an incidence of approximately 6 per million per year in Europe and the USA (Jager et al., 2020; Virgili

et al., 2007). Patients with large choroidal melanomas or melanomas located close to the optic nerve may be treated with enucleation or proton beam therapy (PBT) (Bensoussan et al., 2016; Fuss et al., 2001; Maschi et al., 2016; Weber et al., 2015). These two treatment modalities are equivalent with regard to oncological

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outcomes (i.e. risk of metastases and mortality) (Mosci et al., 2012). However, the options differ with respect to ocular preservation, treatment procedures and side effects, such as change in visual acuity. The most appropriate treatment for a particular patient will therefore depend on how the patient evaluates aspects other than the oncological outcomes. In the process of making a treatment decision, patients will need to consider the information most relevant to their decision. The physician plays a pivotal role in providing accurate and comprehensive information. It may be overwhelming to a patient if the physician discloses all possible information about the treatment modalities (Eppler & Mengis, 2004; Hibbard & Peters, 2003; Longo et al., 2010). It is therefore preferable to give key information that will assist the patient in the decision-making process. To date, it remains unclear what exactly patients diagnosed with choroidal melanoma consider important when faced with the decision between enucleation and PBT.

The aim of this study was to determine the most important treatment characteristics, including outcomes, when choosing between enucleation and PBT. We further aimed to determine to what extent relevant characteristics and outcomes are routinely mentioned during consultations with newly diagnosed choroidal melanoma patients who face the choice between enucleation and PBT.

2 | METHODS

2.1 | Study design

This is a cross-sectional study in which a Swing Weighting (SW) survey was developed and administered, and real-life consultations were audio recorded and analysed. The SW survey was used to assess which treatment characteristics and outcomes are most important to choroidal melanoma patients when choosing between enucleation and PBT. Consultations between newly diagnosed choroidal melanoma patients and their ocular oncologist were studied to assess which treatment characteristics and outcomes are mentioned in current practice. Hereafter, we define 'treatment characteristic' as any aspect related to the treatment procedure itself and its outcomes. The Medical Ethical Committee of the Leiden University Medical Center (LUMC) approved this study (P17.195). All participants provided written informed consent. This study is registered at www.clinicaltrials.gov (identifier: NCT05377957).

2.2 | Research setting

All eligible patients were recruited at the LUMC, a Dutch academic hospital and national referral centre for ocular malignancies. Approximately 200–250 patients per year are diagnosed with uveal melanoma at the LUMC (Netherlands Cancer Registry, 2023). Large and juxtapapillary choroidal melanomas are treated with enucleation at the LUMC or with PBT at the Holland Particle Therapy Center (Holland PTC) in Delft according to a

local protocol (Marinkovic et al., 2021; Rasch et al., 2023; Roelofsen et al., 2021). The choice between enucleation and PBT is offered to patients with a tumour prominence or basal diameter exceeding 7 or 16 mm, respectively, or in case of a tumour with >1/3 involvement of the optic disc (juxtapapillary). Patients treated for a local tumour recurrence or with metastasis at primary diagnosis were not eligible for the present study.

2.3 | Swing weighting survey

The SW survey development consisted of three steps (Figure 1). We conducted (1) a comprehensive literature search followed by (2) interviews to determine all characteristics that differentiate between enucleation and PBT. (3) We conducted an online ranking survey to select the most relevant characteristics. Different (patient) populations were involved in each step; all patients had been diagnosed with choroidal melanoma. The SW survey (Monzani et al., 2021; Whichello et al., 2023) was used to investigate how patients weigh the importance of the seven most important characteristics for deciding between enucleation and PBT.

2.3.1 | Development of the swing weighting survey

Literature search

We conducted the literature search in PubMed (Table S1; literature search string). The search yielded 687 articles. Based on title and abstract, 70 articles were selected that described patients with choroidal melanoma with tumour prominences >7 mm or a juxtapapillary localization, who had been followed up for a median of at least 5 years. After full-text screening, 16 articles describing characteristics which differ between enucleation and PBT in the patient population of interest were included (Affeldt et al., 1980; Bensoussan et al., 2016; Dendale et al., 2006; Egan et al., 1989; Egger et al., 2003; Gragoudas et al., 1982, 1987; Lane et al., 2011; Marinkovic et al., 2021; Maschi et al., 2016; Mosci et al., 2012; Papakostas et al., 2017; Petrovic et al., 2014; Schonfeld et al., 2014; Starr, 1962; Thariat et al., 2016). Eventually, 11 unique characteristics were identified in these 16 articles.

Interviews

Semi-structured individual interviews with patients and medical experts, including a physicist, a nurse, ocular oncologists ($n=4$) and radiation oncologists ($n=4$), were conducted to identify any additional characteristics. The patients ($n=11$) were recruited at the LUMC and had been treated with PBT ($n=5$), enucleation ($n=5$) or PBT followed by a secondary enucleation ($n=1$). The patients were asked about their reason for choosing a particular treatment, their treatment experience and what they consider important for other patients to know prior to deciding about treatment. The medical experts were asked about their years of experience, what patients should know about the treatment options and how they think the treatment options

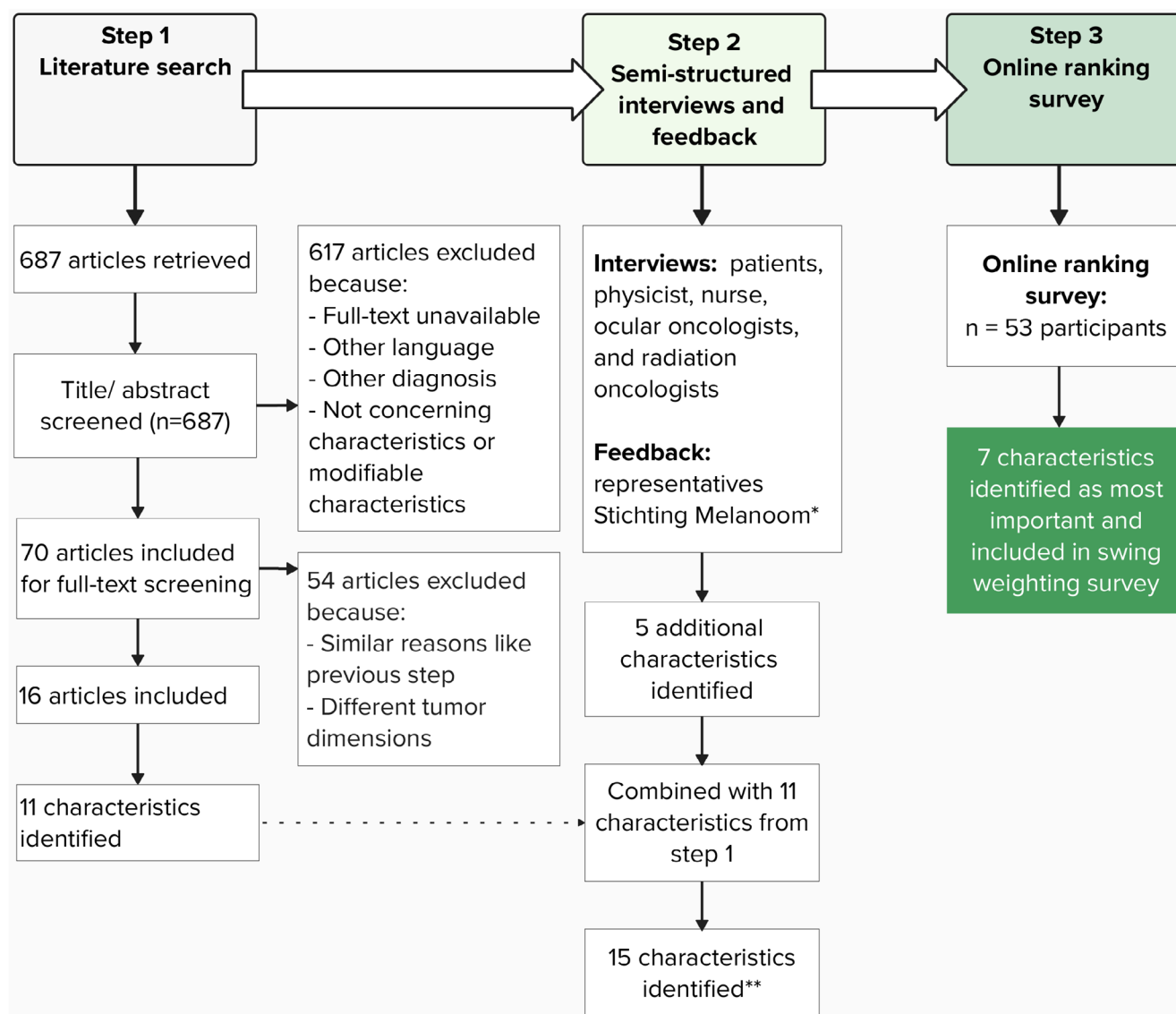


FIGURE 1 Overview of the process to identify the characteristics that differentiate enucleation from PBT to be included in the swing weighting survey. *Stichting Melanoom is the Dutch Melanoma Patient Foundation. **Two characteristics were combined into one.

affect patients' lives. The list of treatment characteristics resulting from the literature was then presented to all participants, and they were asked to propose additional characteristics.

The lists of characteristics resulting from the literature and interviews were aggregated. Overlapping characteristics were combined and renamed when appropriate, resulting in 15 characteristics that differentiate between enucleation and PBT. The definitions of the characteristics were subsequently discussed with four representatives of the Dutch Melanoma Patient Foundation (Stichting Melanoom) to verify and improve clarity (definitions provided in Table S2).

Online ranking survey

We aimed to limit the number of characteristics for inclusion in the SW survey to seven, to provide the patients with a manageable task when asked to consider them. To this end, an online ranking survey was conducted among treated choroidal melanoma patients. The survey was built with QUALTRICS XM software and distributed through the online platform of the Dutch

Melanoma Patient Foundation. Participants were asked to select their top five most important characteristics out of the 15 characteristics, regarding the choice between enucleation and PBT. Additionally, they were asked to rate their top five characteristics between 1 (least important) and 10 (most important). A total of 53 participants (treated with: enucleation, $n=14$, PBT, $n=18$, ruthenium brachytherapy or stereotactic radiotherapy, $n=21$) responded to the ranking survey. The results are provided in Table S3.

2.3.2 | Swing weighting survey

For the SW survey, patients treated with Ruthenium brachytherapy, PBT or enucleation, and patients yet to undergo treatment were included. They were asked to rank the seven characteristics in order of importance for the hypothetical scenario that they could improve (swing) these characteristics from the worst to the best outcome (level) (Figure 2). The levels were based on outcomes reported in previous studies or, if not reported in

Characteristics	Most important
Injection in the eye You will get injections → you will not get any injection	1. Tumor recurrence 10 out of 100 → 3 out of 100
Tumor recurrence 10 out of 100 → 3 out of 100	2. Decrease in vision 100 out of 100 → 56 out of 100
Pain and/or discomfort 25 out of 100 → 0 out of 100	
Other visual complaints 38 out of 100 → 0 out of 100	
Loss of eye 100 out of 100 → 10 out of 100	
Decrease in vision 100 out of 100 → 56 out of 100	
Treatment course 1 operation and 4 radiation treatments, 10-15 consultations → 1 operation, 2-3 consultations	
	Least important

FIGURE 2 Visual representation of the first part of the SW survey—the swinging. An example in which two characteristics are already swung to the right. The worst (in red) and best (in green) levels are shown under the characteristic. Tumour recurrence (worst level, probability of 10 out of 100, best level, probability of 3 out of 100) is ‘swung’ first, and decrease in vision (worst level, probability of 100 out of 100, best level: Probability of 56 out of 100) is ‘swung’ second. The remaining five characteristics should thereafter also be swung to the right in the patient’s order of importance.

the literature, determined based on the cohort of patients treated at the LUMC since June 2019. We documented the full range of outcomes specified for our patient population and used the most extreme reported outcomes, which represented the best or the worst outcome for a specific characteristic. Participants were then asked to assign points (weights) between 0 and 100 to indicate the importance that they assigned to these improvements for each characteristic. The first, and therefore most important characteristic, was automatically assigned the highest weight of 100. The remaining characteristics were presented to patients in the order of importance in which the patient had ranked them in the swinging step. They could assign points to the remaining characteristics on a scale of 0–100, but they were instructed to allocate the points in descending order as consistently as possible. Patients could go back to the swinging step to rearrange the ranking at any time. The survey was developed with Sawtooth Software’s SSI Web (Sawtooth Software. OREM. UT.US), and a link to the survey was distributed to the patients by email.

2.4 | Assessment of real-life consultations

We audio-recorded consultations with newly diagnosed patients to assess how often the 15 characteristics considered most relevant to differentiate between enucleation and PBT are mentioned in routine practice. At the LUMC, patients are first seen by one of four ocular oncologists (GL, JB, MM and TV) who informs them about their diagnosis and explains the treatment options.

Before this consultation, the physician has already had several interactions with the patient throughout the day and unrelated to treatment decision-making. After the consultation, patients usually also have a consultation with an oncology nurse to discuss more practical aspects of both treatments. When patients want to have more information about or opt for PBT, they are referred to one of two radiation oncologists (CR or PB).

For a subset of patients, we also audio recorded the follow-up consultation with the radiation oncologists (hereafter referred to as ‘paired recordings’). All audio recordings were transcribed verbatim, and transcripts were anonymized. A coding scheme was developed to assess whether the characteristics had been mentioned (Table S5). Physicians participating in the consultations were unaware of the 15 characteristics selected and the content of the coding scheme.

The coding scheme distinguished ‘loss of eye’ as primary treatment (enucleation) from ‘loss of eye after PBT’. It further allowed to code the explicit mention of *eye preservation after PBT*, as opposed to *eye loss after PBT*. Comparably, we distinguished between tumour recurrence and tumour control when coding the treatment characteristic ‘tumour recurrence’. We additionally coded what outcomes were mentioned beyond the actual characteristics alone, that is, outcomes that underly or could cause a characteristic. For example, an additional outcome would be ‘cataract’ which could cause the characteristic ‘decrease in visual acuity’. All transcripts were coded independently by at least two assessors (AS, LV and AP), whereafter any inconsistencies were discussed to reach consensus.

2.5 | Statistical analyses

For the analysis of the SW survey, the weight (range: 0–1) of a characteristic a for an individual participant i ($x_{i,a}$) was calculated, by dividing the points the participant had allocated to that characteristic by the total number of points the participant had allocated. Subsequently, the average weight of a characteristic across the entire sample $\bar{S}(a)$ was calculated by the sum of the individual weights of a characteristic for all participants divided by the total number of participants (n) (16), using the following equation: $\bar{S}(a) = \frac{1}{n} \sum_{i=1}^n \frac{x_{i,a}}{\sum_{j=1}^7 x_{i,j}}$. The association between the two most important characteristics and factors that may influence patients' responses was investigated using multiple linear regression for each characteristic separately. The dependent variable was the weight of each characteristic ($x_{i,a}$). Patient age (continuous), sex and actual (previous or planned) treatment for uveal melanoma (Ruthenium brachytherapy, PBT or enucleation) were all entered as independent variables in each model. Statistical significance was accepted for $p < 0.05$ using a two-sided test. Descriptive statistics, such as frequencies, were used to report patient characteristics and how often characteristics had been mentioned in consultations. Statistical analyses were conducted with spss version 29 (IBM Corporation, Armonk, US).

3 | RESULTS

3.1 | Swing weighting

Between April 2023 and September 2024, a total of 270 participants were invited to participate in the SW survey. One hundred and sixty-five (61%) of them completed the SW. Half of the sample was female, the mean age was 61 years and most had undergone Ruthenium brachytherapy. The patient characteristics are listed in Table 1.

The SW analysis showed that overall, the risk of tumour recurrence (weight, 0.20, SD 0.05) and loss of the eye (weight, 0.18, SD 0.05) was considered the two most important characteristics for treatment decision-making. Eye injections (weight, 0.12, SD 0.05) and course of the treatment (weight, 0.11, SD 0.05) were found least important.

Linear regression analysis showed that the female patients found the risk of tumour recurrence less important (weight, 0.19) than the men (weight, 0.21, odds ratio [OR] –0.019, 95% confidence interval [CI] –0.034 to –0.005; $p = 0.010$).

Enucleation and PBT substantially differ regarding eye loss and course of treatment. Despite these differences, the allocated weight for eye loss was not significantly different between patients who had undergone enucleation (weight, 0.17) and patients treated with PBT (weight: 0.18, OR: –0.018, 95% CI: –0.041 to 0.005; $p = 0.131$).

Similarly, there were no significant differences in weights regarding other treatment characteristics in patients who had undergone enucleation compared to PBT. The weights that the patients assigned to the seven most important treatment characteristics are illustrated in Figure 3.

3.2 | Information provided during consultations

From July 2020 until April 2024, a total of $N = 126$ audio consultations were audio recorded. One hundred and ten consultations were conducted by an ocular oncologist; 16 were follow-up consultations with a radiation oncologist. In all consultations with the ocular oncologist, the clinician first informed the patient about their definitive diagnosis and then provided information about the two treatment options. Sixty-one patients were female, the mean age was 64 years, and 66% were treated with PBT. Patient characteristics are described in Table 1.

Overall, 18–43 consultations were audio recorded per ocular oncologist (GL, JB, MM and TV). The average consultation duration was 13 min (range: 5–35 min). Out of the seven most important characteristics distinguishing enucleation from PBT, 'course of treatment' and 'decrease in visual acuity' were mentioned most frequently, namely in 96% and 85% consultations with an ocular oncologist, respectively. The two characteristics rated in the SW survey as most important, that is, 'tumour recurrence' and 'loss of eye after PBT' (i.e. having had a secondary enucleation), were mentioned in 47%, and 42% of the consultations, respectively (Table 2). As stated earlier, a distinction was made whether 'tumour recurrence' or 'tumour control' was mentioned in the consultations. Either 'tumour recurrence' or 'tumour control' was mentioned in, respectively, 3% and 39% of the consultations. Both terms were mentioned in approximately 5% of the consultations.

During consultations that lasted 15 min or less, 'tumour recurrence' and 'loss of eye after PBT' were mentioned in 42% and 36%, respectively. In consultations

TABLE 1 Patient characteristics.

Characteristics	Patients in swing weighting survey ($N = 165$)	Patients who had consultation audio recorded ($N = 110$)
Female sex (%)	79 (48%)	61 (56%)
Mean age at diagnosis (years)	61 (range: 25–82)	64 (range: 25–89)
Treatment (%)		
Ruthenium	89 (54%)	4 (3%)
Proton beam therapy	53 (32%)	73 (66%)
Enucleation	22 (13%)	33 (30%)

Weights of treatment characteristics

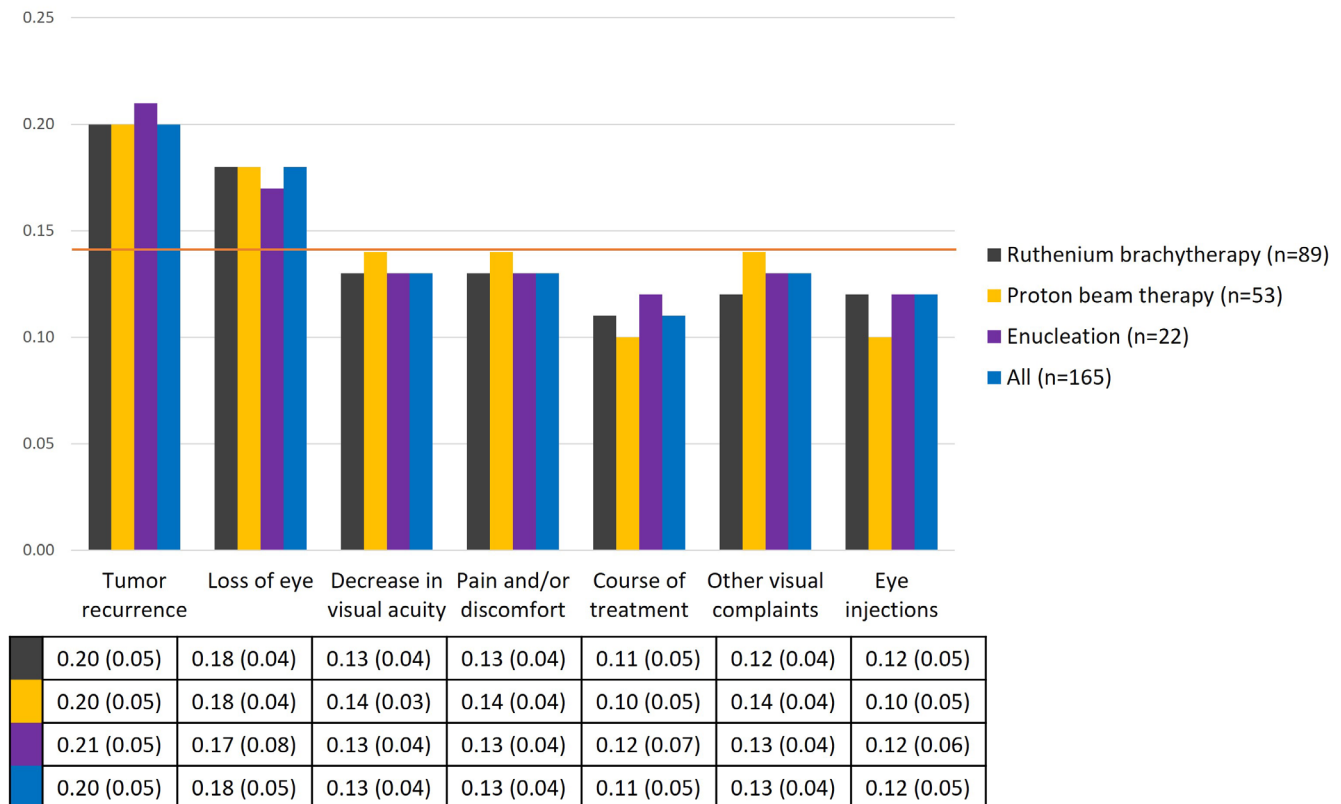


FIGURE 3 Overview of the treatment characteristics differentiating enucleation from PBT, and their respective weight. The standard deviation (SD) of the weight is given in parentheses. The orange line is the mean weight of all characteristics, 0.14.

that lasted more than 15 min, these two important characteristics were both mentioned in almost 60% of the consultations (Table S6). On average, the ocular oncologists mentioned three out of the seven (range: 0–7) most important characteristics, and five out of 15 (range: 1–10) relevant characteristics per consultation. In addition, the ocular oncologists mentioned a total of 14 additional treatment outcomes, such as cataract, retinal detachment or optic nerve damage, in 76 out of 110 consultations (Table S4). Additional treatment outcomes were mentioned 183 times across the 110 consultations. The ocular oncologists explicitly linked 130 of the 183 (61%) additional treatment outcomes to one of the 15 treatment characteristics; in other cases, the relevant treatment characteristic was not mentioned during the consultation.

Sixteen follow-up consultations were audio recorded with the radiation oncologists. These consultations lasted 34 min on average (range: 26–44 min). In these consultations, five out of the seven (range: 4–6) most important characteristics were mentioned on average, and nine out of 15 (range: 7–12) relevant characteristics.

How frequently ocular and radiation oncologists mentioned the 15 characteristics distinguishing enucleation from PBT is listed in Table 2. In the sample of paired recordings, seven out of the 15 characteristics were mentioned by either the ocular oncologist or radiation oncologist. ‘Tumour recurrence’, ‘decrease in visual acuity’, ‘pain and/or discomfort’, ‘course of treatment’, ‘eye injections’, ‘physical changes of the eye/eyelids’ and ‘frequency of follow-up consultations’ were mentioned in the 16 consultations by either one of the two physicians.

Furthermore ‘loss of eye after PBT’ was mentioned in seven consultations. ‘Course of treatment’ and ‘decrease in visual acuity’ were the most-frequently mentioned characteristics in both consultations. The list of frequencies of the paired consultations is listed in Table S7.

4 | DISCUSSION

This is the first study that identified what treatment characteristics, including the treatment outcomes, patients diagnosed with large or juxtapapillary choroidal melanoma consider most important for deciding between enucleation and PBT, and what treatment information physicians routinely provide about the characteristics that differ between the two treatment modalities. Of the 15 characteristics identified as important to distinguish enucleation from PBT, patients found local tumour recurrence and losing their eye after PBT most important. These two characteristics were mentioned in 47% (tumour recurrence) and 42% (eye loss) of the consultations. We found that ocular oncologists mentioned ‘course of treatment’ and ‘decrease in visual acuity’ most frequently with patients. Characteristics such as ‘phantom vision’ and ‘use or need of eye drops’ after treatment were mentioned least frequently.

Risk of local tumour recurrence was considered most important in making the treatment decision. How important a characteristic is considered in decision-making, is closely related to the extent to which the characteristic may potentially vary. The range used for ‘tumour

TABLE 2 Frequency with which ocular and radiation oncologists mentioned treatment characteristics distinguishing enucleation from PBT during consultations by treatment it was related to, if applicable.

Characteristic in order of importance ^a	How often distinguishing characteristic is mentioned			
	Ocular oncologists (N = 110 consultations)		Radiation oncologists (N = 16 consultations)	
	%	95% confidence interval	%	95% confidence interval
Tumour recurrence	47	46–48	94	92–97
After enucleation only	–			
After PBT only	36			
After each treatment	11			
Secondary enucleation	42	36–38	56	50–62
Loss of eye after PBT	26	24–28	25	20–30
Preservation of eye after PBT	18	17–19	50	44–56
Decrease in visual acuity	85	84–86	100	100–100
After enucleation only	2			
After PBT only	59			
After each treatment	24			
Pain and/or discomfort	9	9–10	100	100–100
After enucleation only	1			
After PBT only	8			
After each treatment	–			
Course of treatment	96	96–96	10	100–100
After enucleation only	2			
After PBT only	38			
After each treatment	56			
Other visual complaints	10	10–10	6	3–9
After enucleation only	4.5			
After PBT only	4.5			
After each treatment	1			
Eye injections after PBT	54	5–6	94	91–97
Physical changes of the eye or eyelids	41	40–42	100	100–100
After enucleation only	28			
After PBT only	6			
After each treatment	7			
Tumour regression	28	27–29	81	76–86
After enucleation only	1			
After PBT only	27			
After each treatment	–			
Additional ocular operations	30	32–34	69	63–75
After enucleation only	–			
After PBT only	30			
After each treatment	–			
Frequency follow-up consultations	75	74–76	100	100–100
After enucleation only	2			
After PBT only	45			
After each treatment	28			
Dry and/or watery eye	4	4–5	13	9–17
After enucleation only	1			
After PBT only	3			
After each treatment	–			
Requirement of eye drops after PBT	0		63	57–69
Phantom vision	1	0.7–1	0	–

TABLE 2 (Continued)

Characteristic in order of importance ^a	How often distinguishing characteristic is mentioned			
	Ocular oncologists (N = 110 consultations)		Radiation oncologists (N = 16 consultations)	
	%	95% confidence interval	%	95% confidence interval
After enucleation only	1			
After PBT only	–			
After each treatment	–			
Maintenance of the ocular prosthesis	1	0.7–1	6	3–9

Note: Definition of abbreviation(s): CI = confidence interval.

^aTreatment characteristics 1–7, importance based on SW survey; treatment characteristics 8–14, importance based on rating survey.

recurrence' in the SW survey was based on the published range of probabilities of local recurrence after enucleation 3%–5% (Shammas & Blodi, 1977; Starr, 1962), and after PBT (3%–10%) (Riechardt et al., 2020; Weber et al., 2015). Given the clinically relevant range presented in probability of tumour recurrence, these findings show that this outcome is important to patients facing the treatment decision.

Our study suggests that patients also consider eye preservation as very important. This may be interpreted as either a desire to preserve or not to preserve the affected eye. The desire to preserve one's eye may be driven by the wish to keep as much vision as possible, especially in those with impaired vision of the contralateral eye. Besides, cosmetic reasons or not wanting to lose an important organ can be a reason to consider eye preservation important, even if the eye has lost its functional capacity. The latter is supported by the observation described in Marinkovic et al. (2021), 5% of the patients in the study were already legally blind (visual acuity <0.1) (Social Security, 2017) in the affected eye, but still made the effort to travel abroad to undergo PBT to spare their affected eye. On the other hand, there may also be patients who explicitly want their affected eye to be removed. Reasons may be to get rid of the tumour mass, to minimize the risk of local recurrence, a shorter treatment and follow-up and/or that treatment has lower risks of side effects. However, earlier research has shown that worry about tumour recurrence does not differ between patients who have undergone enucleation or PBT (Hope-Stone et al., 2019).

Interestingly, our study did not show a significant difference in how important patients treated with enucleation versus PBT considered eye loss. This suggests that the influence of personal treatment experience on the response to the hypothetical scenario of the SW survey is limited. It should be noted that this study was not powered for this sub-analysis; small differences between treatment groups cannot therefore be ruled out.

Local recurrence risk was determined to be most important for the treatment decision, and one might expect that especially those who had undergone enucleation would rate eye loss as substantially less important than recurrence risk. Our data show, however, that the difference in importance between local recurrence risk and eye loss is small. This implies that most patients, including those who underwent enucleation, find information on both the risk of recurrence and eye loss important.

It is remarkable that patients did not consider the course of treatment very important. A primary enucleation consists of a day admission with just one follow-up visit, while the PBT regimen is a time-intensive trajectory that involves an operation for the placement of clips, five visits to the proton therapy centre and frequent follow-up visits at the hospital in the first years. Seemingly, outcomes of treatment may be more important to patients than treatment procedures.

Furthermore, we investigated which of the 15 characteristics considered important ocular oncologists routinely provide during the consultation in which the patient receives their diagnosis and learns about the two treatment options. The ocular oncologists mentioned the two characteristics that patients consider most important, that is, risk of tumour recurrence and loss of eye after PBT, in 47% and 37% of the consultations, respectively. Ocular oncologists most often mentioned information about the treatment procedures, such as course of treatment (96%) and frequency of follow-up consultations (75%) and decrease in visual acuity (85%). Research both within (Kunnenman et al., 2015; Snijders et al., 2014) and outside (Burger et al., 2019; Ghulam et al., 2006; Knops et al., 2010; Teshome et al., 2018) of oncology showed that relevant information such as information about benefits and harms is not always provided to patients. Physicians mentioned probability of survival after cancer in less than 20% of consultations with patients diagnosed with rectal cancer (Kunnenman et al., 2015) and left frequently occurring and impactful side effects of treatment of rectal and sigmoid cancer (e.g. faecal incontinence) often unsaid (Snijders et al., 2014). In addition, patients also report missing information from their physician when discussing cancer, such as cancer recurrence (Ankersmid et al., 2023; Keegan et al., 2012).

Given the limited time available during consultations, and in order not to overwhelm patients, the results of the present study can help prioritize what information ocular oncologists could provide to patients facing the treatment decision. Ideally, the physician would inform patients at least about the seven most important treatment characteristics. These characteristics may not need to be discussed in detail, except if patients ask for more information.

At our hospital, and following the first consultation with the ocular oncologist, the patient immediately sees the nurse for additional information. We did not record

those conversations and cannot report which additional information they may provide. Nurses usually inform patients about the logistical and administrative aspects of the treatments. In addition, all patients receive information leaflets about the diagnosis and treatment modalities after their consultation with the nurse. In these leaflets, 12 out of the 15 characteristics are mentioned, including the seven most important characteristics. Moreover, if the patient opts for PBT, they are referred to the radiation oncologist for more in-depth information about PBT. These consultations last longer, giving the physician more time to discuss PBT with the patient.

After consulting both physicians, patients who opted for PBT had not been completely informed about all information considered relevant for the decision. However, possibly with additional information the nurse provided and the information leaflets, patients who opted for PBT likely had obtained or at least had access to information about the seven most important characteristics for decision-making. It should be noted that this information is important to all patients making the decision between enucleation and PBT, and not only to those opting for PBT. Patients who had opted for enucleation over PBT did not see a radiation oncologist and therefore had not received all information for decision-making.

In this study, we also documented the additional treatment outcomes ocular oncologists mentioned or explained during the consultations. Given that most patients generally lack medical knowledge to fully understand potential consequences of those treatment outcomes, it would be helpful to communicate the impact of those treatment outcomes explicitly. For instance, when mentioning damage to the retina or a retinal detachment, ocular oncologists may explain that these side effects could cause, for example, a decrease in visual acuity or can lead to additional (surgical) interventions. The ocular oncologist explicitly mentioned the characteristic in relation to the additional outcome in almost 70% of the cases. In other 30% of the times an additional outcome was mentioned, we cannot assume that patients understood how that additional outcome would impact them. Moreover, the possibility of secondary enucleation after PBT can be described as risk of 'loss of eye after PBT' or as 'likelihood of preserving the eye after PBT'. Either description demonstrates the inevitable choice of a particular frame (i.e. negative and positive, respectively). The chosen frame directs attention to, in this case, the negative or the positive outcome (Kühberger, 2023; Martinez et al., 2024). The most neutral manner to describe this outcome would be to state the outcomes using both frames. We recommend to mention both frames when informing the patient about the two most important characteristics, that is, both tumour recurrence and tumour control, and both losing and preserving the eye after PBT.

Ocular oncologists can aim to reduce the provision of information to what is most likely relevant to the patient. This study's top seven characteristics can provide guidance for this selection in consultations with patients diagnosed with choroidal melanoma who are eligible for both enucleation and PBT. We also investigated whether there is a correlation between patient characteristics such as age or sex and the importance of the characteristics,

and cannot recommend tailoring information provision based on these features as we did not find associations. For effective information provision, it may be helpful if the various healthcare providers involved in informing patients determine together who is responsible for providing what information about the treatment modalities and their characteristics.

Further research is needed to determine whether patients understand the information as it is provided about the treatment modalities. It would also be interesting to investigate whether the extent to which they have been informed about the characteristics is related to their satisfaction with the decision that was made (Walming et al., 2018). Lastly, intervention studies are needed to investigate the use and effect of conversation aids to support information provision (Bentley et al., 2021; Brito et al., 2018; Durand et al., 2021), or the implementation of joint consultations with both ocular and radiation oncologists. Both could be strategies to improve information provision and to prepare patients newly diagnosed with choroidal melanoma eligible for enucleation and PBT better for decision making.

4.1 | Limitations

This study has some limitations. First, we sent the SW survey to both newly diagnosed patients and patients who had already been treated, regardless of whether these patients were eligible for Ruthenium brachytherapy, PBT or enucleation to reach a sufficiently large sample in a timely manner. All these patients share the experience of being diagnosed with choroidal melanoma. Those who had or would receive Ruthenium brachytherapy (will) undergo similar treatment procedures and experience similar side effects as patients who received PBT. The fact that part of the patients already had made a treatment choice or even had undergone treatment may have influenced their responses. However, no significant correlation was found between the received treatment and the characteristics. Therefore, any influence on the survey responses is unlikely. In addition, there may have been some selection bias for the audio consultations. Consultations with patients who were expected to be more challenging or emotionally difficult, based on previous interactions with the patient, may have been less likely to have been recorded. In these consultations, information provided about characteristics might have been different from the consultations with the included patients, which could cause an under- or overestimation of the frequencies reported in this study.

5 | CONCLUSION

Our study shows that Dutch patients diagnosed with choroidal melanoma consider local tumour control and eye preservation most important when facing a treatment decision between enucleation and PBT. The information that ocular oncologists routinely provide only partially overlaps with what patients consider important for decision-making. Our results can be used to determine what information to provide when counselling

newly diagnosed patients with choroidal melanoma. Ocular oncologists, radiation oncologists and nurses are encouraged to make deliberate and aligned decisions regarding their information provision in order to support patient decision-making.

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